EXPERIMENT 5

Design a lexical Analyzer to find the number of whitespaces and newline characters.

CODE:

```
#include <stdio.h>

int main() {
   char c;
   int space = 0, newline = 0;

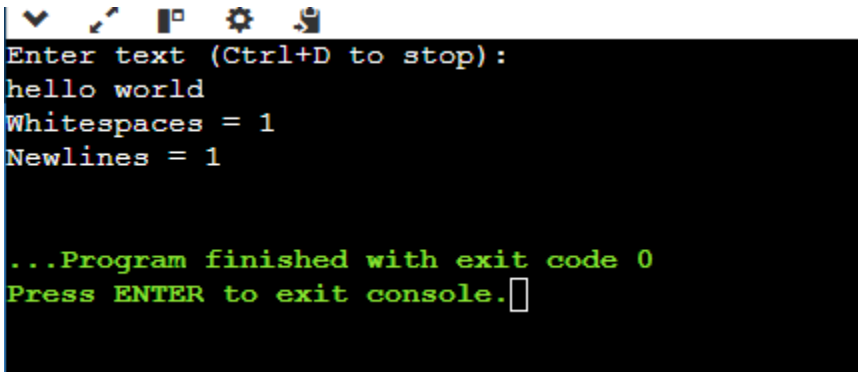
   printf("Enter text (Ctrl+D to stop):\n");

   while ((c = getchar()) != EOF) {
      if (c == ' ' || c == '\t')
         space++;
      if (c == '\n')
         newline++;
   }

   printf("Whitespaces = %d\n", space);
   printf("Newlines = %d\n", newline);

   return 0;
}
```

OUTPUT:



EXPERIMENT 6

Develop a lexical Analyzer to test whether a given identifier is valid or not.

CODE:

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char s[50];
    int i, valid = 1;

    printf("Enter identifier: ");
    scanf("%s", s);

    if (!isalpha(s[0]) && s[0] != '_')
        valid = 0;

    for (i = 1; s[i] != '\0'; i++)
        if (!isalnum(s[i]) && s[i] != '_')
            valid = 0;

    if (valid)
        printf("Valid Identifier");
    else
        printf("Invalid Identifier");

    return 0;
}
```

OUTPUT:

